

Step 4: Future Overpass Forecast

Predict upcoming satellite passes from real recent acquisition patterns — not textbook assumptions.

Rather than assuming each satellite's textbook nominal revisit cycle (10 days for Sentinel-2, 16 for Landsat, etc.), this step looks at the last ~60 days of *actual* acquisitions over your specific AOI, detects the real repeating gap pattern, and projects it forward. Real patterns often differ from the nominal figure — for example, an AOI sitting in the overlap between two adjacent orbit paths gets imaged more often than the nominal cycle implies.

Reading the "basis" column

- "empirical pattern [...]" — a real repeating gap was detected in recent history; trust this.
- "nominal N-day cycle..." — not enough recent history to confirm a real pattern, so it fell back to the textbook figure; treat this as lower-confidence.

Confidence by sensor

- **Sentinel-2, Landsat, NISAR** — acquired systematically over land on every overpass, so these predictions are reliable.
- **MODIS** — near-daily and systematic, so its forecast is basically "tomorrow, and every day after" — correct, but not a very selective planning signal by itself.
- **Sentinel-1** — acquisition is *tasked* by ESA, meaning the satellite flies over on schedule but whether it actually collects data for your AOI on a given pass is a separate decision this can't see. Treat these predictions cautiously, especially with a "nominal" basis.

■ **Tip** — For an independent cross-check that doesn't rely on acquisition history at all, use the Live Tracker step — it predicts geometric overpasses directly from orbital mechanics.